

Building an Azure Cloud Security Detection Lab with Microsoft Sentinel: A Hands-On Guide



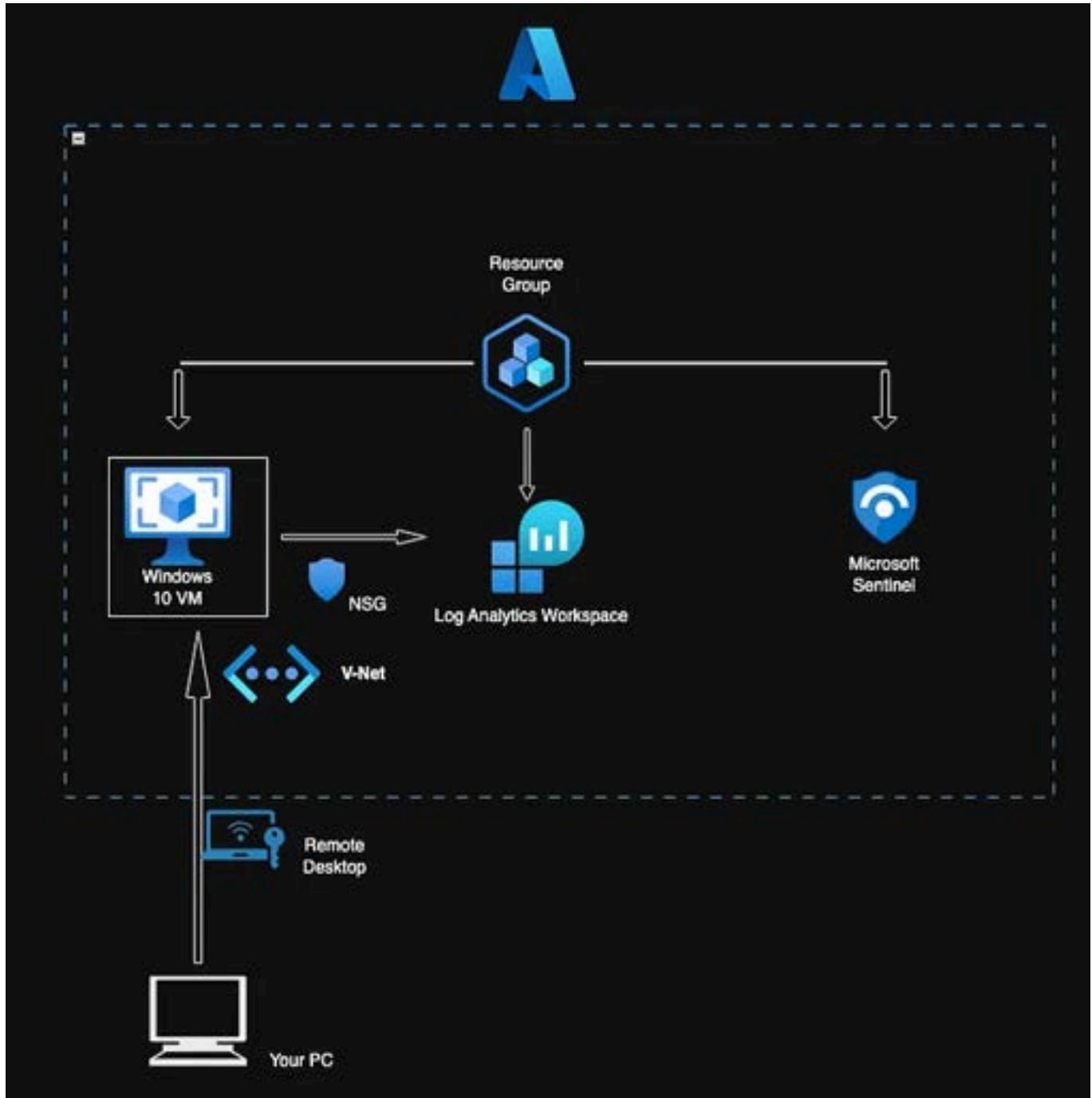
Eric Clark · 6 min read · 4 hours ago



This documentation covers my hands-on experience setting up a Virtual Machine in Azure, establishing resources like Microsoft Sentinel, importing logs into Sentinel through data connectors, implementing Azure security measures, and building custom logs with Kusto Query Language (KQL).

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A High Level Overview



Environment Architecture Diagram

Lab Objectives:

- Set up and launch Azure components, including Log Analytics Workspace, Virtual Machines, and Azure Sentinel
- Apply industry-standard security practices for networks and virtual machines
- Use Data Connectors to import information into Sentinel for examination
- Comprehend Windows Security Event logging systems
- Set up Windows Security Policies
- Apply KQL for log queries
- Create custom Analytic Rules for identifying Microsoft Security Events
- Leverage MITRE ATT&CK framework to identify adversary strategies and detection approach.

Azure Components Utilized:

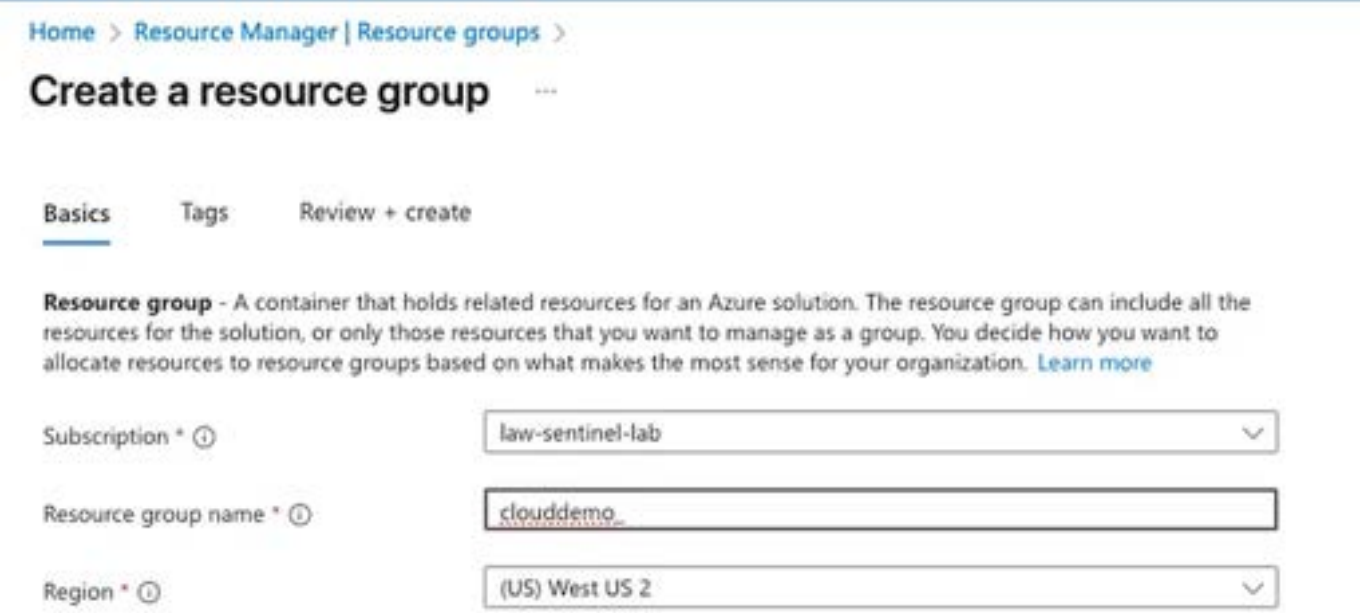
- Windows 10 Virtual Machine
- Microsoft Sentinel
- Log Analytics Workspace

Let's Get Started:

I started by establishing an Azure account through <https://azure.microsoft.com/en-us/free/>. Microsoft offers \$200 in complimentary credits for hands-on practice, which is ideal. However, it's critical to remove these resources after completion to preserve your remaining credits.

My first task involved establishing a Resource Group (abbreviated as RG throughout this guide). RGs function as organizational containers for Azure

solutions or related resources you wish to manage. Creating one was straightforward: I navigated to 'Resource Group' under Navigate → Create and established a basic RG named clouddemo_, as shown below.



Home > Resource Manager | Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

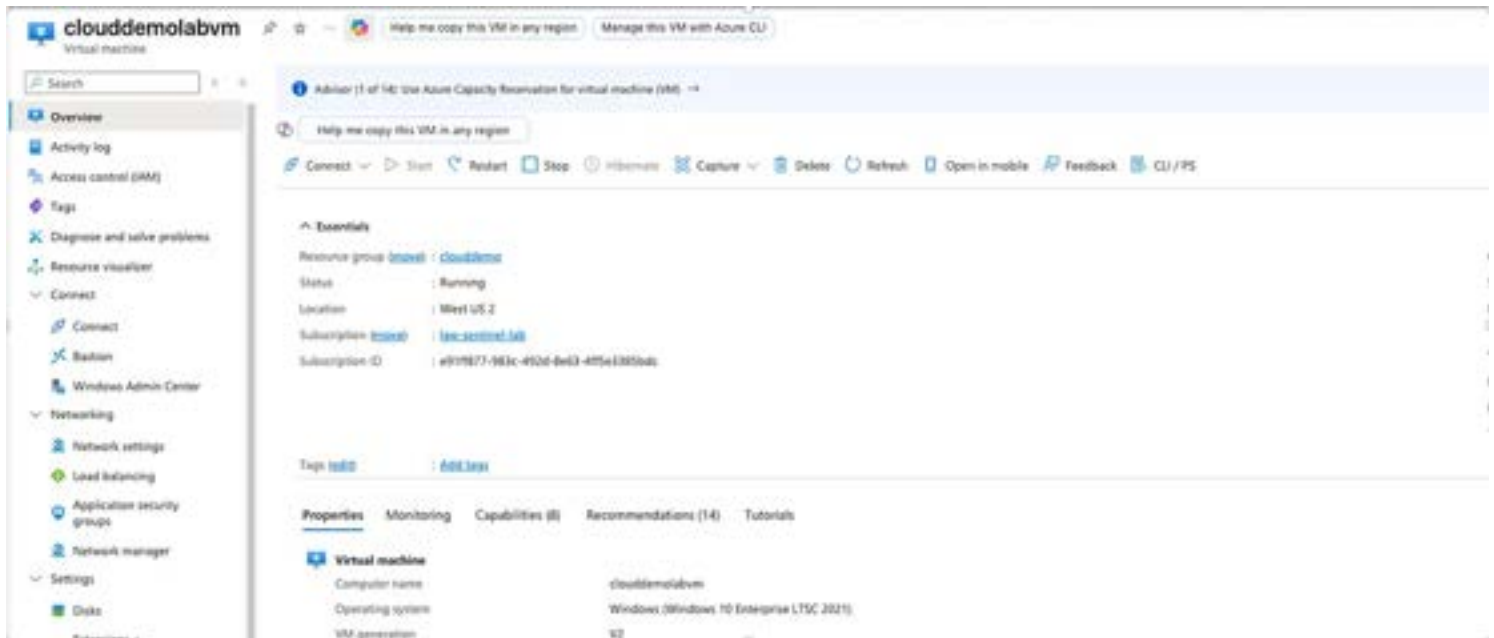
Subscription * ⓘ law-sentinel-lab

Resource group name * ⓘ clouddemo

Region * ⓘ (US) West US 2

Resource Group Creation

Following this, I deployed a Virtual Machine that would serve as my data collection source. My process included: Selecting the Virtual Machine tab → Create, choosing my previously created RG, naming my VM, selecting the Windows 10 Enterprise image, establishing login credentials (essential for subsequent steps), then clicking Review + Create while maintaining default settings for everything else.



During VM deployment, a Network Security Group (NSG) was automatically generated. NSGs function similarly to firewalls for Azure resources, controlling traffic permissions to and from the virtual network. Upon VM deployment, it connects to the virtual network (vnet) and receives an IP address along with a network interface. The NSG contains security rules specifying destination, source, port, and protocol parameters.



Remote Desktop Protocol (RDP) is configured by default for VM access. The NSG displayed that all RDP traffic was permitted, which presents significant security concerns. Threat actors commonly scan for exposed management ports like RDP and SSH to compromise machines as initial entry points for further exploitation.

Home > clouddemolabvm-nsg Network security group

Diagnose connectivity issues related to this security group | Retrieve detailed information for troubleshooting security rules | Analyze security rules for this

Search | Move | Delete | Refresh | Give feedback

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Resource visualizer | Settings | Monitoring | Automation | Help

Essentials

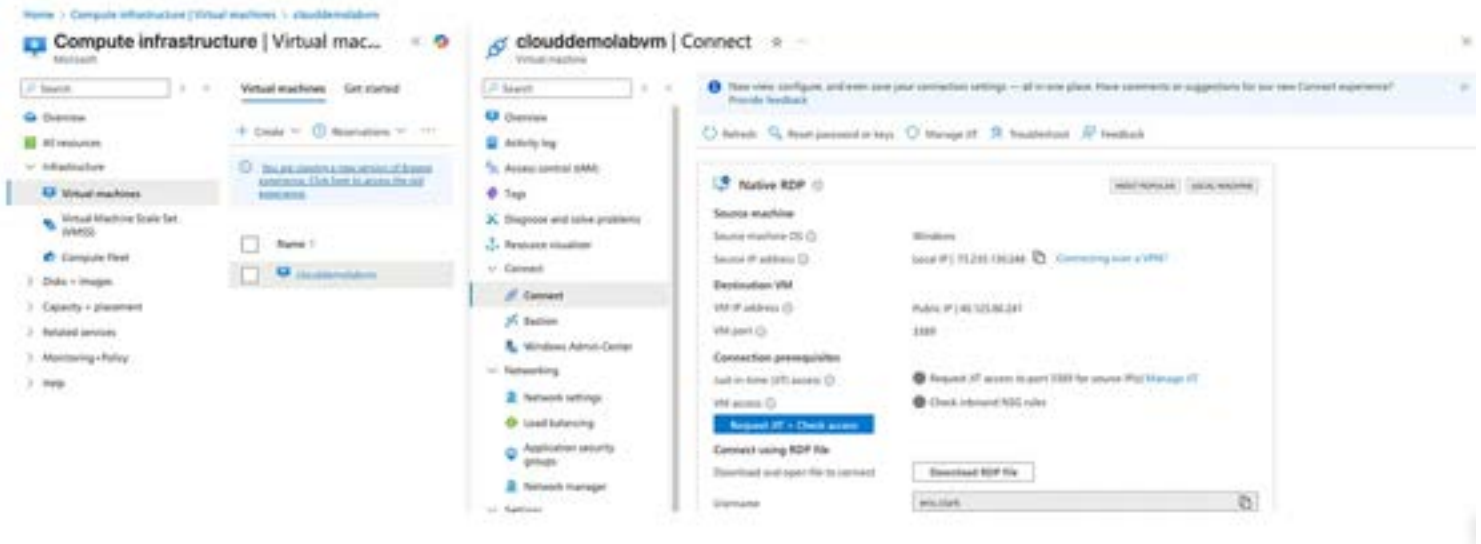
Resource group (choose) : clouddemo Custom security rules : 2 inbound, 0 outbound
 Location : West US 2 Associated with : 0 subnets, 1 network int
 Subscription (choose) : lee-sentinel-lab
 Subscription ID : e91f877-981c-492d-8e63-4f5e3381bdc
 Tags (edit) : Add tags

Filter by name | Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓
Inbound Security Rules				
1000	MicrosoftDefenderForCloud-B	22,3389,5985,5986	Any	Any
1001	RDP	3389	TCP	Any
65000	AllowVnetInbound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalancerInbound	Any	Any	AzureLoadBalancer
65500	DenyAllInbound	Any	Any	Any

RDP currently is open and needs to be restricted to my device by setting up JIT access.

An additional security layer is necessary to restrict RDP access exclusively to my personal device. This can be accomplished by activating Just-In-Time (JIT) access. When these above conditions exist, JIT is suggested. I located and selected Microsoft Defender for Cloud in the Azure Portal, accessed Environmental Settings from the left panel, clicked my Azure Subscription, then enabled all available plans. To activate JIT, I returned to the Defender main page, selected Workload Protections from the left panel, chose Just-in-time VM access under Advanced Protections, selected my VM, then clicked Enable JIT on VM. I maintained default configurations for management port access, which includes a 3-hour time restriction:



Request JIT (blue icon)

After checking the VM's Network tab, I confirmed that RDP now exclusively permitted my IP address, and JIT is properly configured on my VM:

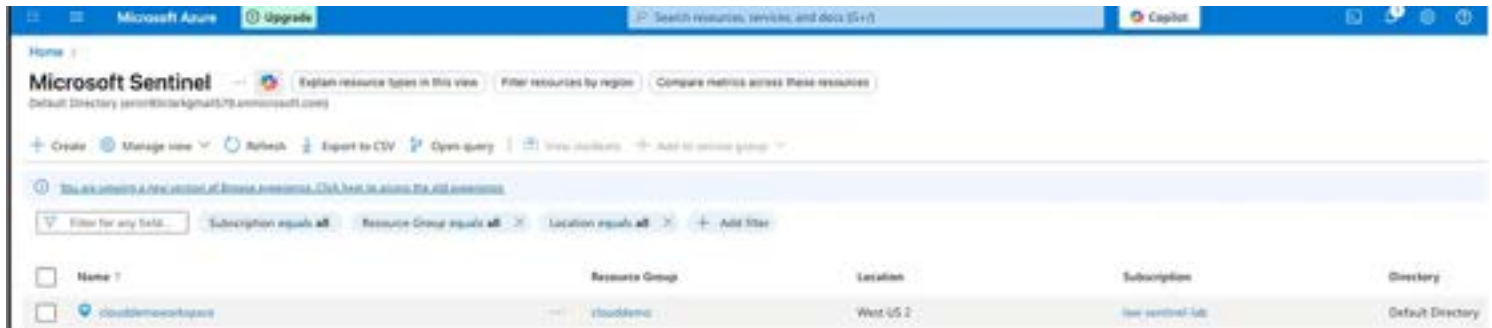


If JIT is successfully configured on your VM, it will appear under “configured” in the JIT VM access section

• • •

Setting Up Log Collection

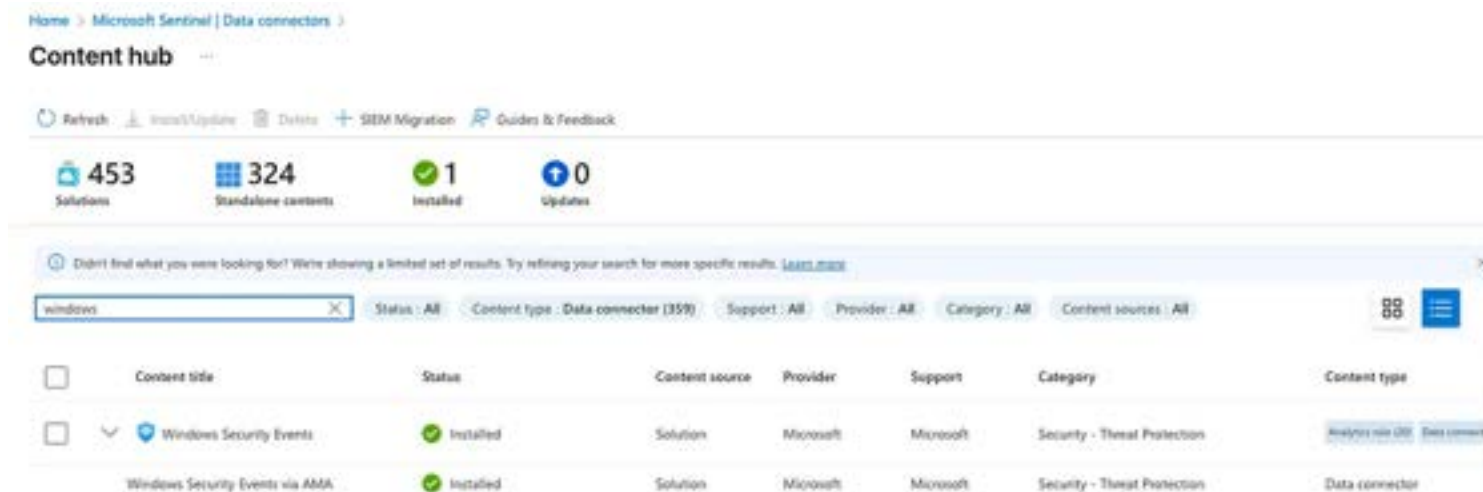
To gather log data, I needed to establish a Log Analytics Workspace within Microsoft Sentinel. My steps included: Searching for Microsoft Sentinel, clicking Create, selecting Create a New Workspace, choosing my RG, naming the workspace 'clouddemoworkspace', then clicking Review + Create.



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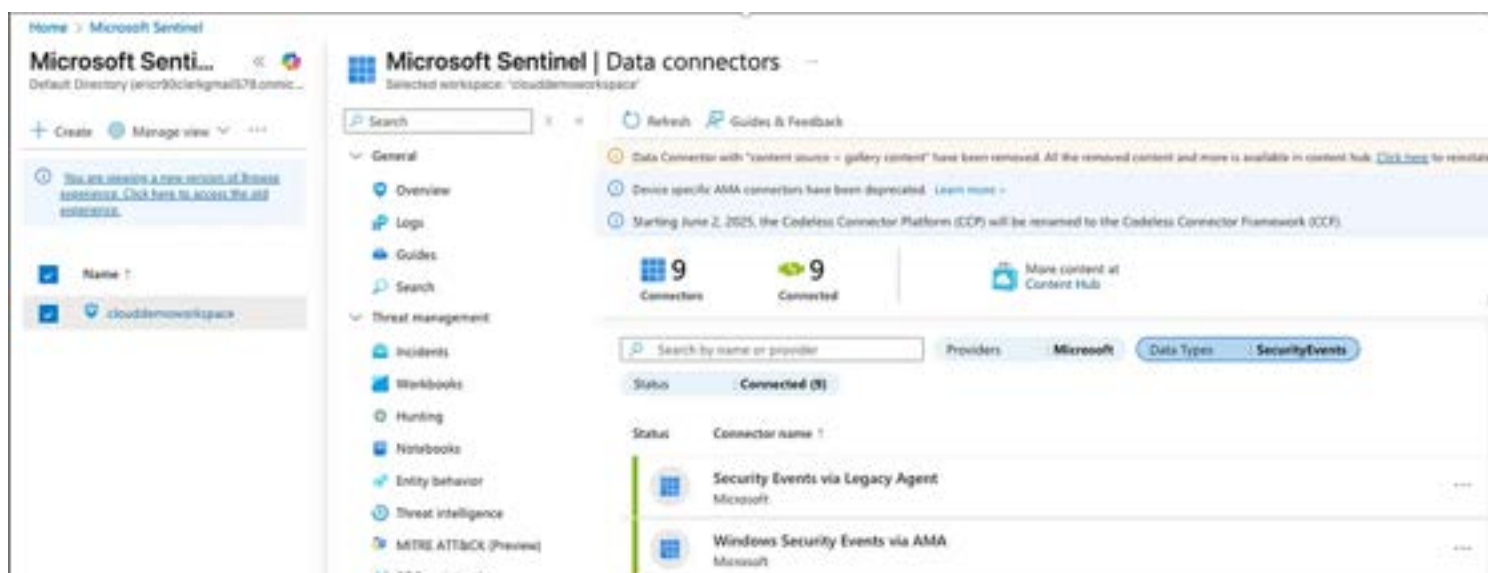
Importing Data into Sentinel

Initially, Sentinel will display no information since no data is being ingested into the logs. The following phase involved implementing data connectors to capture security events that would later be generated from the VM. I clicked Data Connectors, selected More content, searched for security events, chose Windows Security Events, then clicked Install. This installation process adds Windows Events via AMA, which is the connector enabling users to “view dashboard, create custom queries and improve investigation” according to its description. *Please be advised- in order to locate ‘Windows Security Events’, you must search for it within the content hub (pictured below).*



Windows Security Events AMA in Content Hub

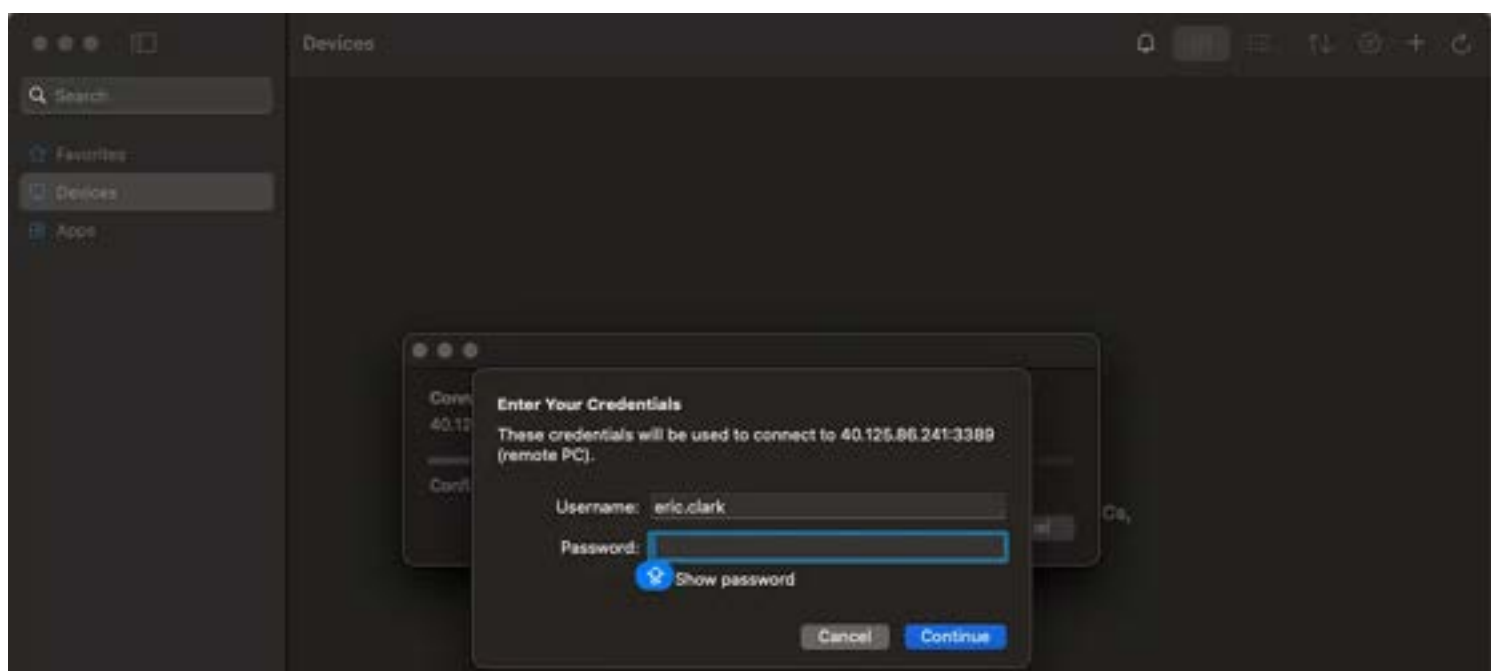
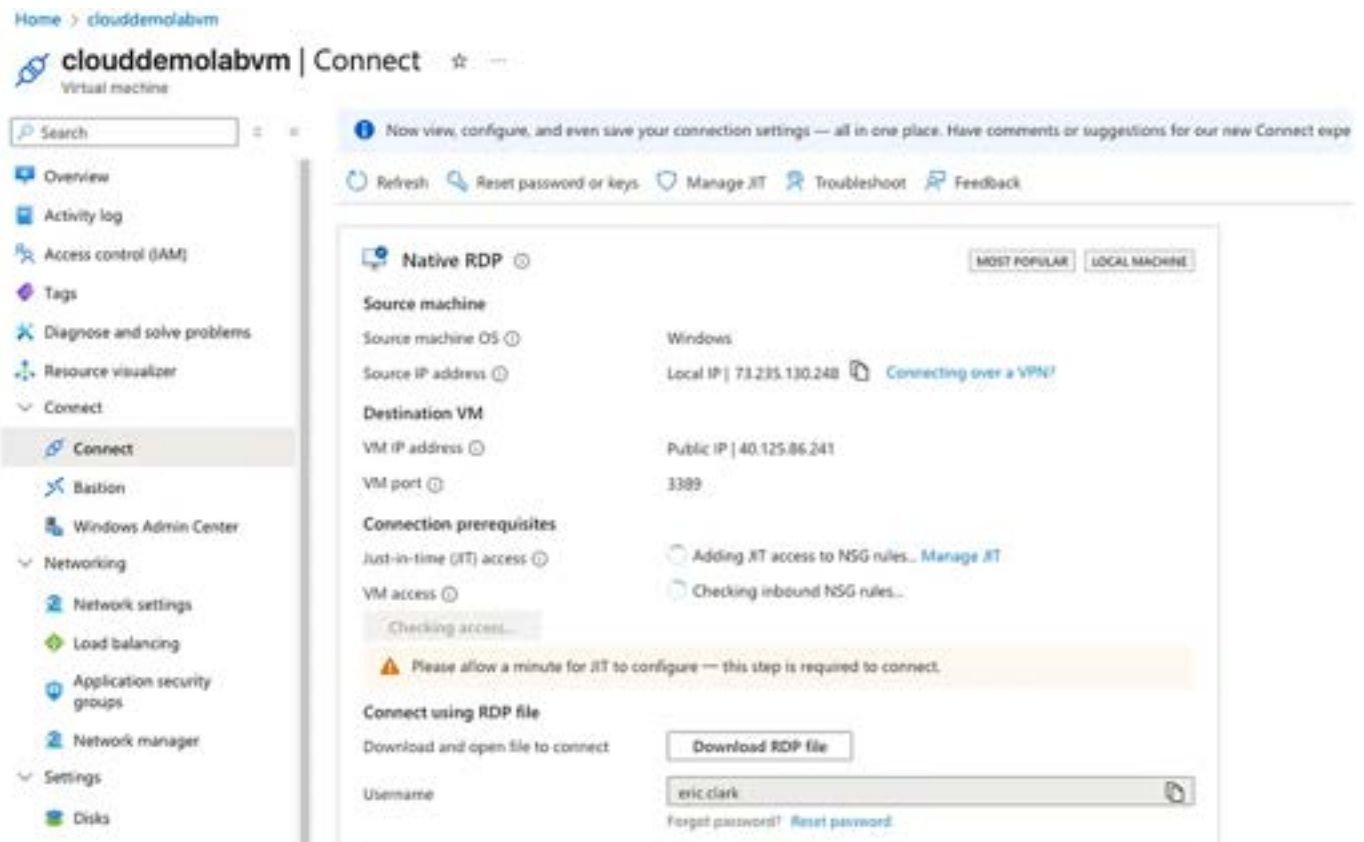
Below, we can confirm data collection rule completion. Note that Windows Sec Events via AMA is listed as a data connector that is filtered as connected to my workspace in Sentinel:



Creating Security Events in the Virtual Machine

The next phase involved generating activity within the VM. Windows maintains records of events like login attempts and processes through

Security Logs in Event Viewer. First, I established an RDP connection to the VM. My steps: I accessed Virtual Machine homepage, clicked Connect, selected My IP under Source IP, then clicked Request Access. With JIT enabled, I needed to request the 3-hour access window for my IP address:



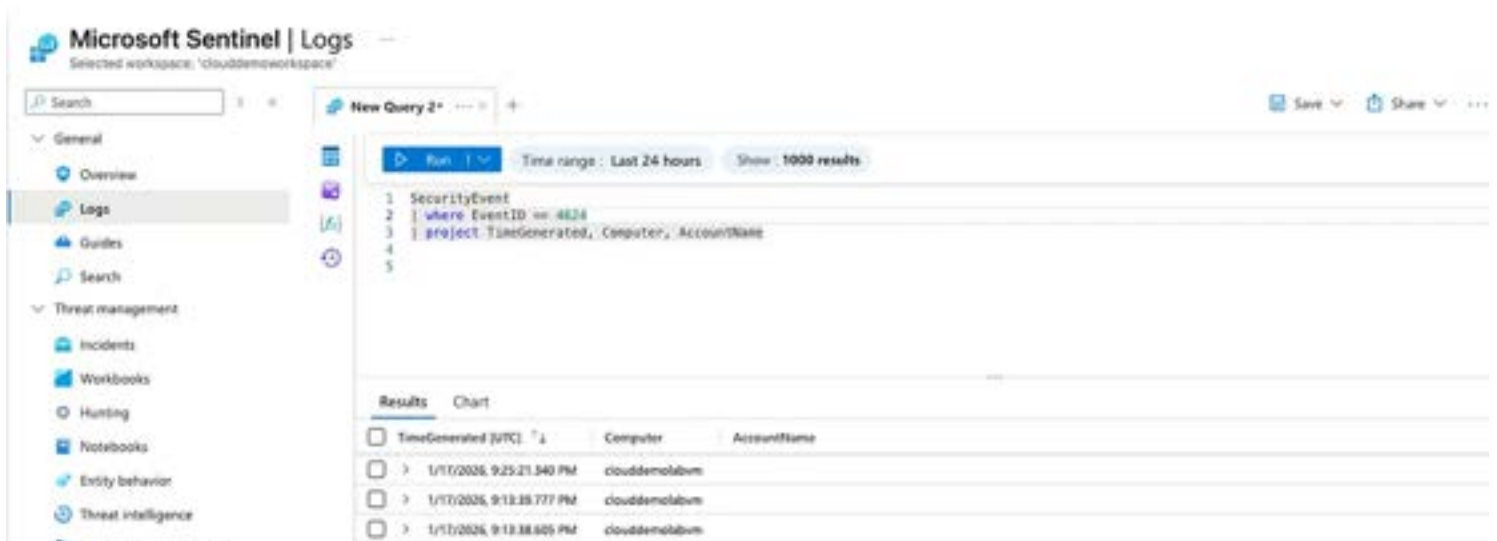


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Developing KQL Scripts

I progressed from generating security events to retrieving event IDs, to employing KQL for generating associated logs. From the Sentinel homepage, I selected Logs from the left panel, clicked in the “Type your query here” field, and entered the KQL logic shown below:

```
SecurityEvent  
| where EventID == 4624  
| project TimeGenerated, Computer, AccountName
```



Log Results after query

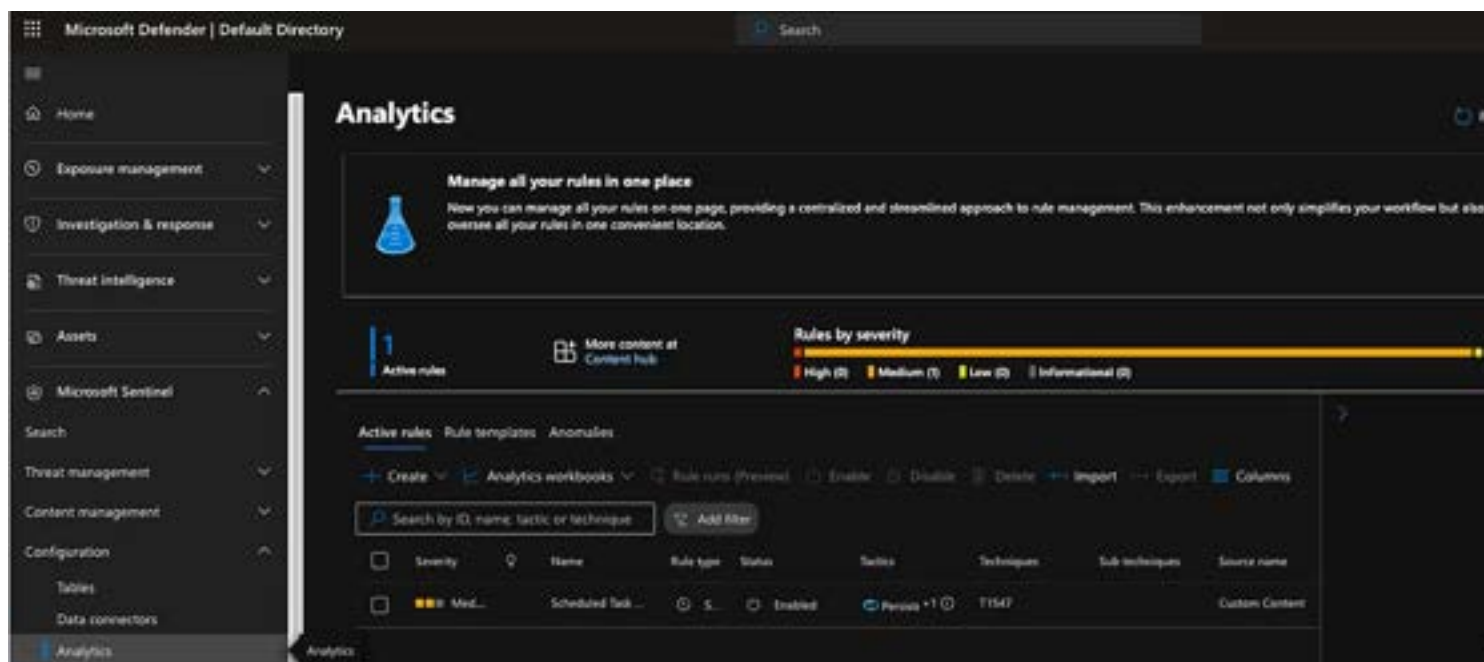
The query searched for security events matching event ID '4624'(Successful logon). This generated columns displaying my login time and the machine used. In an enterprise setting with multiple endpoints, the AccountName field would differentiate users and their logins, but this wasn't necessary for my lab.

Next, I used KQL to build the analytics rule required for generating scheduled task alerts.

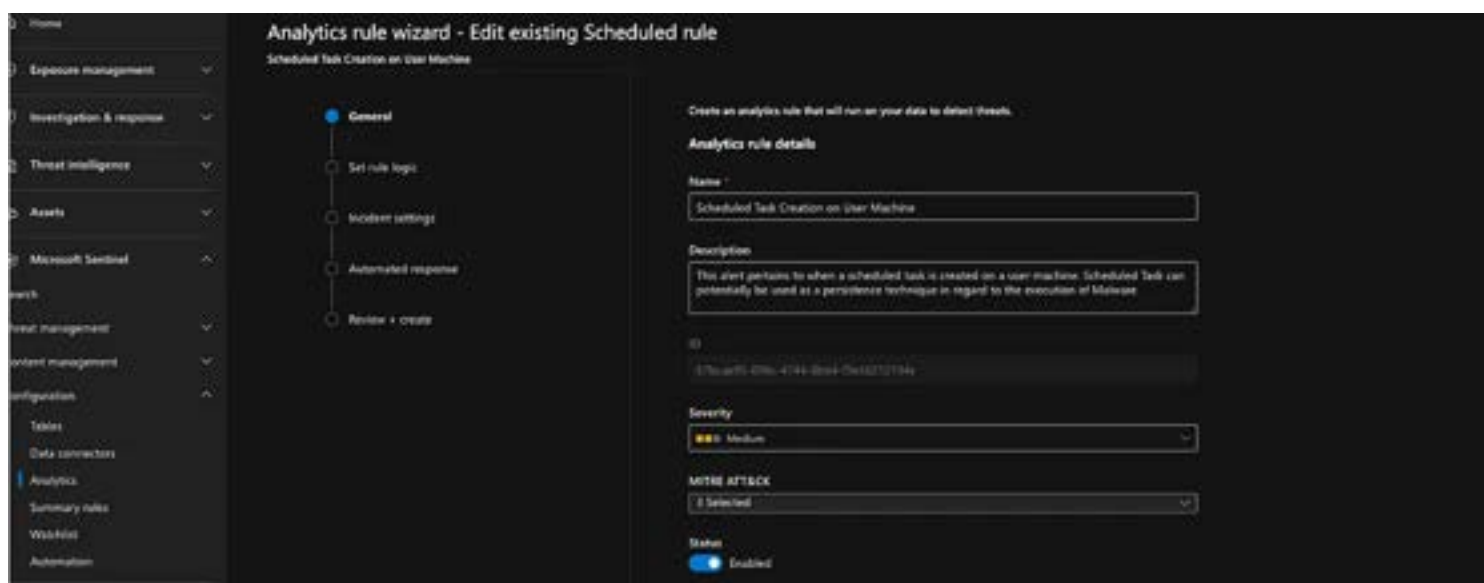
Note: (After selecting Analytics, Azure will redirect you to Microsoft Defender, where the scheduled tasks alerts tool in Sentinel now resides)

My steps:

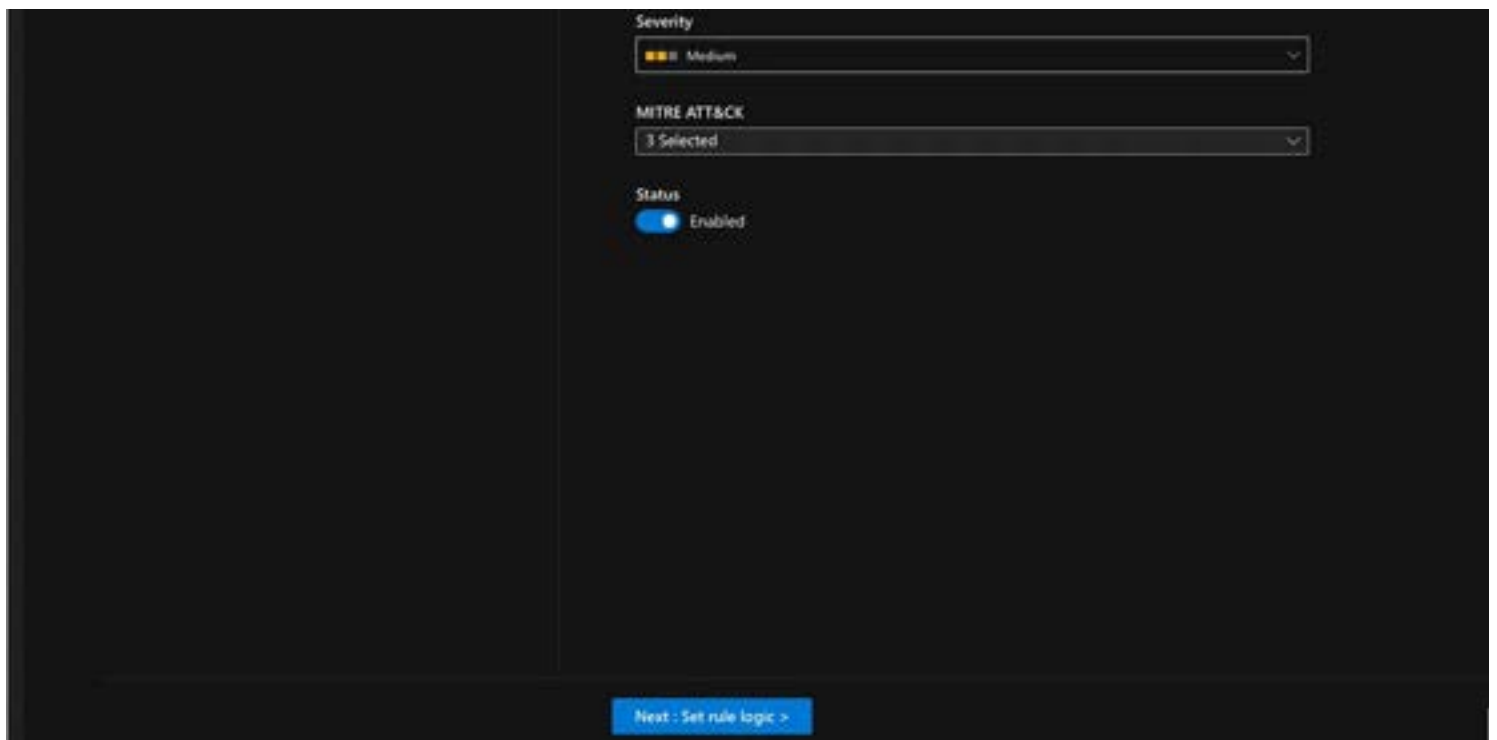
From the Defender homepage, I selected Microsoft Sentinel → Configuration → Analytics:



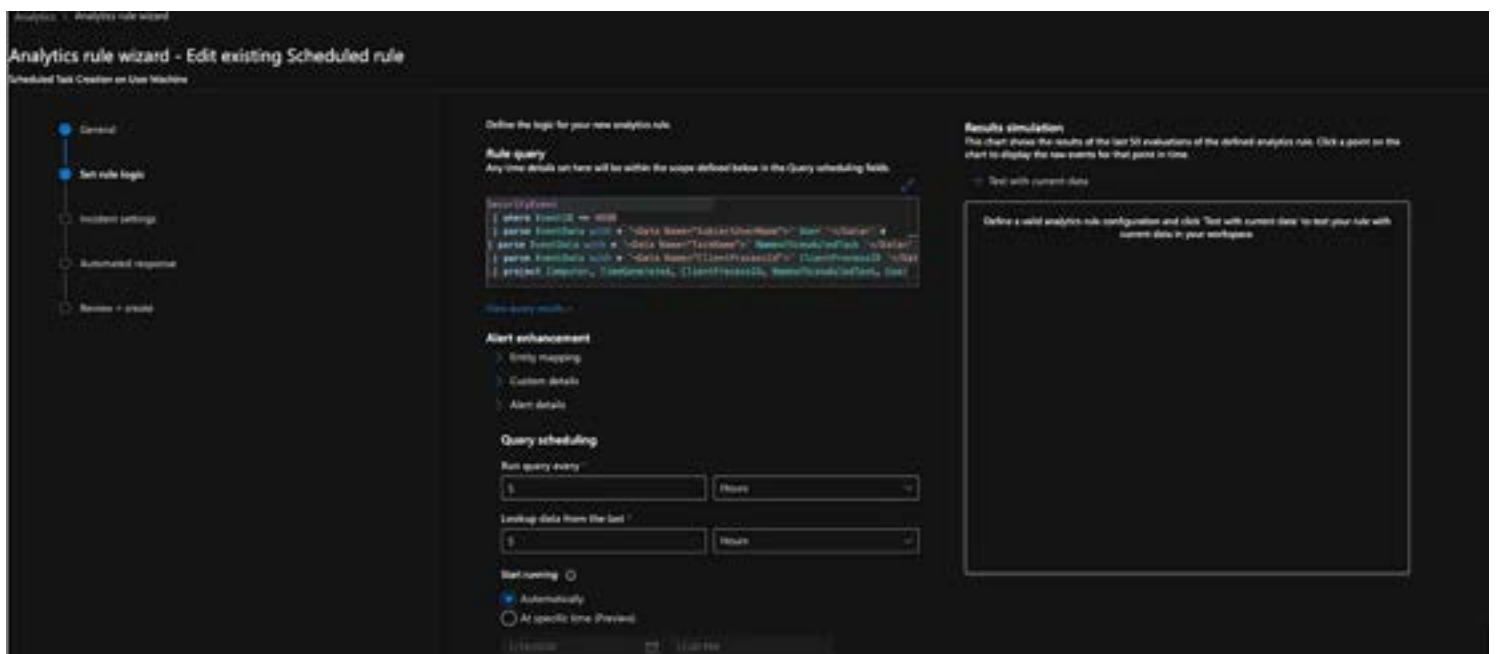
You're then directed to the Analytics rule wizard. Select Create, choose Scheduled Query rule, in this case it's named "Scheduled Task Creation on User Machine", write a description, and select your severity. Be sure to map the MITRE ATT&CK TTP(Tactics, Techniques, and Procedures), and then click next to set your logic :



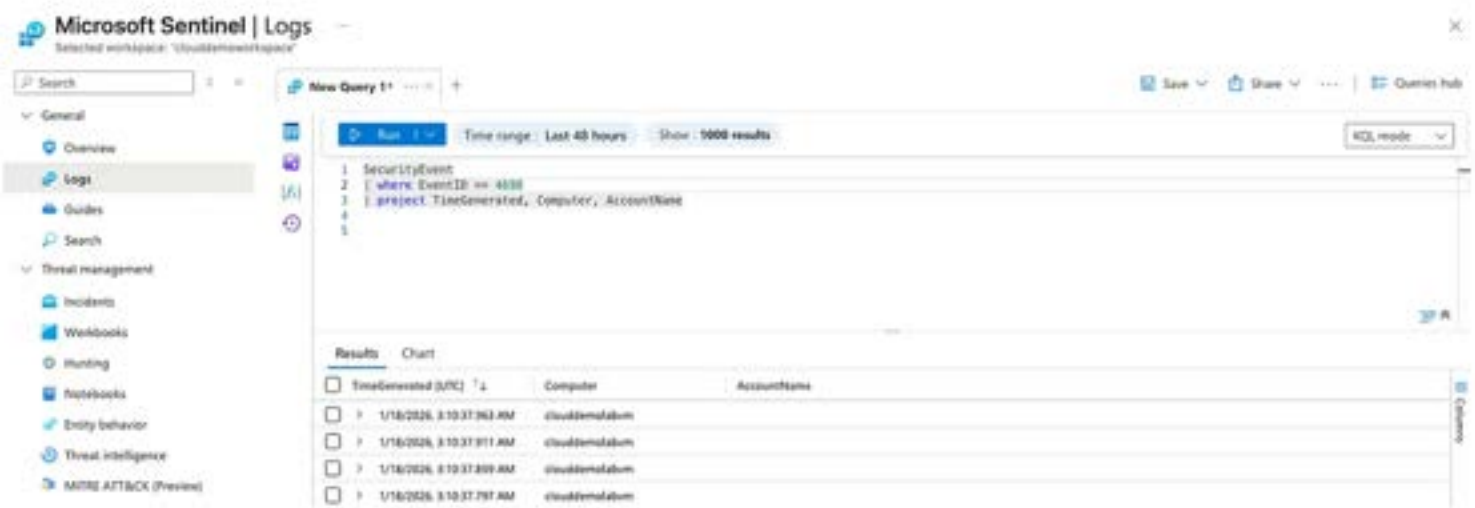
Analytics Rule Wizard



At this stage, we write the query. Go ahead and bypass Incident Settings and Automation response, then click Review + Create and Save:



The scheduled task successfully ran at the designated time. In Sentinel, I received the corresponding alerts generated from Event ID 4698:



This experience provides deeper understanding of how tools like Sentinel and Defender are deployed to monitor and secure cloud resources. Mapping TTPs associated with the scheduled task event was particularly engaging. When applying MITRE ATT&CK framework knowledge, I chose Persistence as the tactic since adversaries can execute malicious programs as scheduled tasks to establish initial or continued system access and deploy additional malicious payloads.

Resource Cleanup:

Finally, I deleted and disabled these resources to preserve my remaining credits. That concludes this lab.

Acknowledgement: This lab was inspired by and follows the excellent work of CyberWox.

Thank you for reading!!!

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